

Car Connect (oops)

Authentication Service

```
package service;

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import dao.DatabaseContext;
import entity.Admin;
import entity.Customer;
import exception.AuthenticationException;

public class AuthenticationService {

    // Customer Login Method
    public boolean login(String username, String password) {
        String query="SELECT customerID, password FROM customer WHERE username = ?";

        try (Connection conn = DatabaseContext.getConnection();
            PreparedStatement stmt = conn.prepareStatement(query)) {
            stmt.setString(1, username);
            ResultSet rs = stmt.executeQuery();
            if (rs.next()) {
                String storedPassword = rs.getString("password");
                if (storedPassword.equals(password)) {
                    System.out.println("Login successful for customer: " + username);
                    return true;
                } else {
                    throw new AuthenticationException("Invalid password.");
                }
            }
        }
    }
}
```

```

        }
    } else {
        throw new AuthenticationException("User not found.");
    }

} catch (Exception e) {
    System.out.println("Authentication error: " + e.getMessage());
}

return false;
}

//Admin Login Method
public boolean loginAdmin(String username, String password) {
    String query="SELECT adminID, password FROM admin WHERE username = ?";

    try (Connection conn = DatabaseContext.getConnection();
        PreparedStatement stmt = conn.prepareStatement(query)) {
        stmt.setString(1, username);
        ResultSet rs = stmt.executeQuery();
        if (rs.next()) {
            String storedPassword = rs.getString("password");
            if (storedPassword.equals(password)) {
                System.out.println("Login successful for Admin: " + username);
                return true;
            } else {
                throw new AuthenticationException("Invalid password.");
            }
        } else {
            throw new AuthenticationException("User not found.");
        }
    }
}

```

```

    }

    } catch (Exception e) {
        System.out.println("Authentication error: " + e.getMessage());
    }
    return false;
}

```

```

// Customer Registration Method

public boolean register(Customer customer) {
    String checkQuery= "SELECT customerID FROM customer WHERE username = ?"; //
    SQL query checks if a username already exists

    String insertQuery= "INSERT INTO customer (firstName, lastName, email,
    phoneNumber, address, username, password, registrationDate) VALUES
    (?, ?, ?, ?, ?, ?, ?, ?)"; //inserts a new customer record

    try (Connection conn = DatabaseContext.getConnection());
        PreparedStatement checkStmt = conn.prepareStatement(checkQuery);
        PreparedStatement insertStmt = conn.prepareStatement(insertQuery)) {
        // Check if username exists
        checkStmt.setString(1, customer.getUsername());
        ResultSet rs = checkStmt.executeQuery();
        if (rs.next()) {
            throw new AuthenticationException("Username already taken.");
        }
        // Insert new customer
        insertStmt.setString(1, customer.getFirstName());
        insertStmt.setString(2, customer.getLastName());
        insertStmt.setString(3, customer.getEmail());
    }
}

```

```
        insertStmt.setString(4, customer.getPhoneNumber());
        insertStmt.setString(5, customer.getAddress());
        insertStmt.setString(6, customer.getUsername());
        insertStmt.setString(7, customer.getPassword()); // Storing plain
password
        insertStmt.setString(8, customer.getRegistrationDate());
```

```
        int rowsInserted = insertStmt.executeUpdate();
        if (rowsInserted > 0) {
            System.out.println("Customer registered successfully.");
            return true;
        } else {
            System.out.println("Customer registration failed.");
        }

    } catch (Exception e) {
        System.out.println("Registration error: " + e.getMessage());
    }
    return false;
}
```

//Admin Registration Method

```
public boolean registerAdmin(Admin admin) {
    String checkQuery = "SELECT adminID FROM admin WHERE username = ?"; // Check
if username already exists

    String insertQuery = "INSERT INTO admin (firstName, lastName, email,
phoneNumber, username, password, role, joinDate) VALUES (?, ?, ?, ?, ?, ?, ?, ?)"; //
Insert new admin
```

```
try (Connection conn = DatabaseContext.getConnection();
    PreparedStatement checkStmt = conn.prepareStatement(checkQuery);
```

```

        PreparedStatement insertStmt = conn.prepareStatement(insertQuery)) {

    // Check if username exists
    checkStmt.setString(1, admin.getUsername());
    ResultSet rs =checkStmt.executeQuery();
    if (rs.next()) {
        throw new AuthenticationException("Username already taken.");
    }

    // Insert new admin
    insertStmt.setString(1, admin.getFirstName());
    insertStmt.setString(2, admin.getLastName());
    insertStmt.setString(3, admin.getEmail());
    insertStmt.setString(4, admin.getPhoneNumber());
    insertStmt.setString(5, admin.getUsername());
    insertStmt.setString(6, admin.getPassword()); // Plain password
    insertStmt.setString(7, admin.getRole());
    insertStmt.setString(8, admin.getJoinDate());

    int rowsInserted =insertStmt.executeUpdate();
    if (rowsInserted > 0) {
        System.out.println("Admin registered successfully.");
        return true;
    } else {
        System.out.println("Admin registration failed.");
    }

} catch (Exception e) {
    System.out.println("Registration error: " + e.getMessage());
}

```

```

    }
    return false;
}
}

```

Main.java

```

Customer customer1 = new Customer(
6,
    "John",
    "Doe",
    "john.doe@example.com",
    "9876543210",
    "123 Main Street, New York, NY",
    "john_doe",
    "securePass123",
    "2024-04-01"
);

```

```

AuthenticationService a=new AuthenticationService();
a.login("kirthidevi", "Abc@1234");

```

```

<terminated> Main (2) [Java Application] C:\Users\WELCOME\.p2\pool\plugins\org.eclipse
[✓] Database Connected Successfully!
Login successful for customer: kirthidevi

```

```

a.login("priyakani", "secure@99");

```

```

<terminated> Main (2) [Java Application] C:\Users\WELCOME\.p2\pool\plugins\org.eclipse.justj.open
[✓] Database Connected Successfully!
Authentication error: Invalid password.

```

```

a.loginAdmin("rahulverma", "Rahul@123");

```

```

<terminated> Main (2) [Java Application] C:\Users\WELCOME\.p2\pool\plugins\org.eclipse
[✓] Database Connected Successfully!
Login successful for Admin: rahulverma

```

```

[X] Database Connected Successfully!
Registration error: Username already taken.

```

```
Admin(5,"kavin","kumar","kavin1125@gmail.com","9876768612","kevin25","Kksen@2325","Super admin","2025-04-05");
a.registerAdmin(a1);
<terminated> Main (2) [Java Application] C:\Users\WELCOME\.p2\pool\plugin:
☒ Database Connected Successfully!
Admin registered successfully.
```

[illegible]